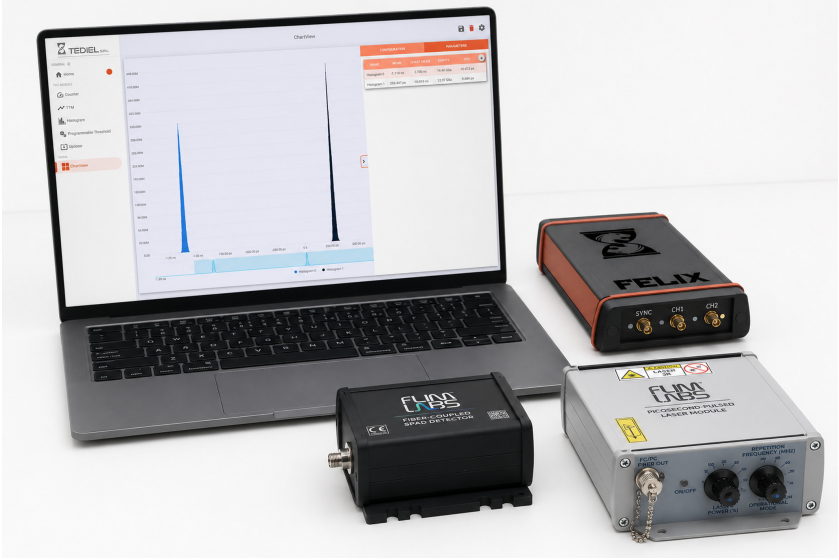
User Manual

# Quantum Photonics Bundle

**QPB-1C1L**




User Manual – Version v1.1.0
30 Oct 2026

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# 1 General Information

This chapter provides essential information on the purpose of this manual, the manufacturer, regulatory compliance, warranty conditions, and the conventions used throughout this document.

## 1.1 About This Manual

This manual provides instructions for the installation, configuration, operation, maintenance, and disposal of the Quantum Photonics Bundle. The bundle provides a compact, PC-controlled single-photon timing platform for quantum optics, photon statistics, time-correlated measurements, quantum photonics education, and related laboratory applications.

The Quantum Photonics Bundlecombines a picosecond pulsed laser source, single-photon SPAD detector options, a high-resolution time-to-digital converter, software, optical fibers, and the required connection accessories.

Individual components included in the bundle may be supplied with their own dedicated manuals containing additional safety, installation, and operating information. These manuals must also be read and followed before installation and operation.

This manual forms an integral part of the product and must remain accessible to the personnel responsible for the system. If the system is transferred to a third party, all accompanying documentation must be delivered with it.

This manual contains proprietary information and may not be reproduced or distributed, in whole or in part, without prior written consent of the manufacturer.

The components of the bundle configurations are listed in Section 2.3. Depending on the ordered configuration, the wavelength of the laser source and the detector model may vary.



**Warning** Before performing any operation, the entire manual and the documentation of all supplied components must be read carefully and the relevant sections reviewed in detail.

### 1.1.1 User Qualification

The bundle is intended for use by qualified personnel familiar with laboratory electronic instrumentation, optical systems, laser sources, single-photon detectors, and time-resolved photon counting measurements.

The system is not a toy and must not be used by children under 14 years of age. Persons with reduced physical, sensory, or cognitive abilities may operate the system only under supervision by qualified personnel.

### 1.1.2 Revision History

Document Revision: v1.1.0 - 30 Oct 2026

A certified copy of this manual is stored in the manufacturer's technical file for the legally required period. During this period, documentation may be requested from the manufacturer.

## 1.2 Symbols and Conventions

This manual includes two different levels of hazard warnings. The warnings may include various safety symbols. The hazard levels and safety symbols comply with the ISO Standard for Product Safety Signs and Labels.

### Hazard Levels



**Warning** Indicates a potential hazard that could result in serious injury or death.



**Caution** Indicates a potential hazard that could result in minor or moderate injury, or equipment damage.

**Notice** Indicates important information that could lead to malfunction, improper integration, or misuse.

### Safety Symbols

The following symbol types can be found on the product or in the manual.

**Table 1.1:** Safety symbols and their meanings

| Symbol | Form | Information |
|---|---|---|
|  | Yellow triangle with a black border and a black symbol | Indicates a hazard. |
|  | Circular sign with a red border, diagonal line, and black symbol | Identifies a prohibited action. |
|  | Blue circle with a white symbol | Identifies a required action. |

**Warning** Safety pictograms performing a protective function must not be removed, covered, or damaged. If a label becomes illegible or damaged, contact authorized service for replacement.

## 1.3 Manufacturer Information

Manufacturer: FLIM LABS S.r.l.

Address: Via della Farnesina, 3 00135 Rome (RM), Italy

Telephone: +39-329 761540

E-mail: info@flimlabs.com

Website: www.flimlabs.com

## 1.4 Declaration of Conformity

The FLIM LABS electronic components supplied as part of the bundle are individually CE-marked where applicable and are supplied with their respective Declarations of Conformity in accordance with applicable European legislation.

The FELIX Time-to-Digital Converter is supplied by TEDIEL S.r.l. and is subject to the regulatory documentation provided by its manufacturer.

The user remains responsible for ensuring that the complete installed system complies with all applicable local regulations, safety requirements, and operating conditions after installation.

## 1.5 Product and Component Labels

The components supplied with the product contain labels with regulatory markings and identification information including:

- Product name and model
- Manufacturer identification
- Regulatory markings where applicable
- Electrical specifications
- Serial number information

The product packaging may additionally contain kit-level identification information including the product number and configuration information.

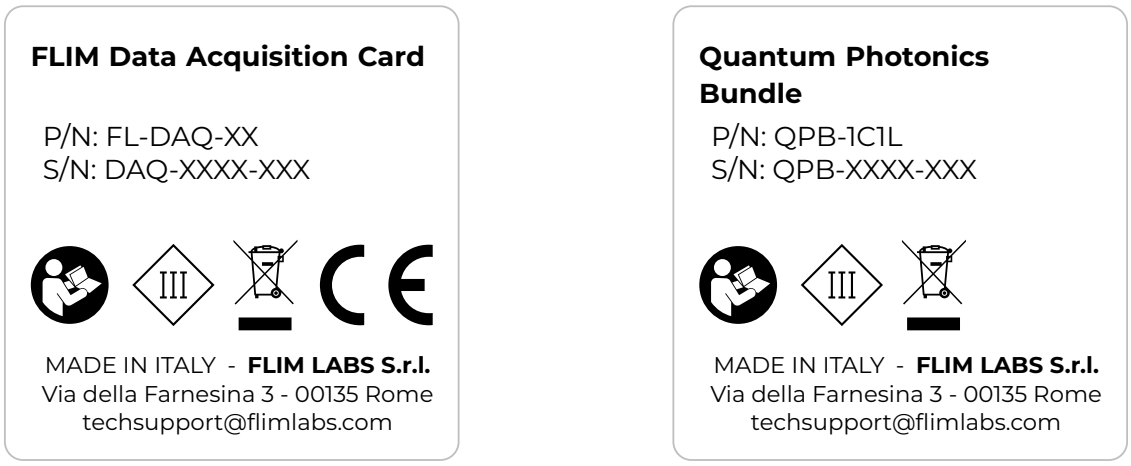



**Figure 1.1:** Product labels for the kit components

Disposal requirements associated with WEEE markings are described in Section 7.2.

## 1.6 Warranty and Liability

### Warranty

The manufacturer provides 24 months warranty for private customers and 12 months warranty for commercial customers. The warranty period begins on the date of delivery.

The warranty covers defects in materials and workmanship under normal and intended use. It is valid only if the system is installed, operated, and maintained according to this manual and the applicable manuals of the supplied components.

Unauthorized modification, improper use, failure to follow instructions, or repair by non-authorized personnel voids the warranty. Full warranty terms are available in the manufacturer's General Terms and Conditions published on the official website.

To submit a warranty claim, provide the following:

- Product type/version
- Proof of purchase, including date
- Detailed description of the issue

### Limitation of Liability

The manufacturer shall not be liable for damages resulting from improper use, including but not limited to:

- Use outside the intended purpose
- Failure to follow this manual or applicable safety instructions
- Installation or operation by unqualified personnel
- Operation outside specified electrical, optical, or environmental limits
- Use of incompatible or non-compliant input signals or connected equipment
- Unauthorized modification, repair, or use of non-original spare parts
- Integration into systems not compliant with applicable safety regulations

Except where otherwise required by mandatory law, the manufacturer shall not be liable for indirect, incidental, or consequential damages, including but not limited to loss of data, loss of profit, or system downtime.

# 2 System Description

This chapter provides an overview of the Quantum Photonics Bundle, its intended application, major system components, and the overall system architecture.

The bundle provides a compact, PC-controlled platform for single-photon timing, photon counting, time-tagging, photon correlation, and hardware histogramming experiments.

## 2.1 Functional Description

The Quantum Photonics Bundleenables time-resolved single-photon measurements using a pulsed optical excitation source, single-photon detection, and high-resolution photon timing electronics.

The system operates by generating short optical pulses with a fiber-coupled picosecond laser module. Photons are detected by a single-photon SPAD detector and converted into electrical timing signals. These signals are processed by the FELIX Time-to-Digital Converter for time-tagging, photon counting, and real-time hardware histogramming.

The connected software provides control, acquisition, visualization, and access to Python and C++ interfaces for automated measurements and custom experimental workflows.

## 2.2 Intended Use

The Quantum Photonics Bundleis intended for laboratory, research, education, and development applications involving quantum photonics, single-photon timing, photon statistics, photon correlation, time-resolved measurements, and related optical experiments.

The bundle is intended exclusively for indoor laboratory use under the environmental and operating conditions specified in this manual and in the manuals of the supplied components.

The bundle is not intended for:

- Medical diagnosis or therapeutic applications
- Operation in explosive or hazardous environments
- Outdoor operation
- Safety-critical applications
- Use as a consumer product

### Reasonably Foreseeable Misuse

Any use outside the intended purpose described in this manual is considered improper use.

Reasonably foreseeable misuse includes, but is not limited to:

- Modification of the supplied hardware or optical components
- Use of incompatible electrical or optical signals
- Bypassing laser safety precautions
- Operation outside specified environmental conditions
- Use without reading the supplied documentation
- Use by unqualified personnel
- Disconnecting optical fibers while the laser is active

Improper use may result in equipment damage, incorrect measurements, reduced performance, or hazardous operating conditions.

## 2.3 System Components Overview

The standard Quantum Photonics Bundlecontains the components listed in Table 2.1. The number of components and exact product numbers depend on the ordered configuration.

## 2.4 System Architecture

Figure 2.1 shows the overall architecture of the Quantum Photonics Bundle.

**Table 2.1:** System components overview

| Component | Part P/N | Description |
|---|---|---|
| Picosecond Pulsed Laser Module | PLD-XXX-SM-FC-U | Fiber-coupled excitation source for time-resolved photon measurements. Available wavelengths include 405, 445, 488, 520, 532, 635, and 850 nm depending on the ordered configuration. |
| Single-Photon SPAD Detector | SPAD-050-C-X-FC | Single-photon avalanche diode detector for photon detection. Blue-shifted and red-shifted detector options are available depending on the required spectral range. |
| FELIX Time-to-Digital Converter | TEBXXXXXX | High-resolution timing electronics for photon time-tagging, correlation measurements, and real-time hardware histogramming. |
| TEDIEL Software Suite | n.a. | Software for TDC control, photon acquisition, visualization, and programmable access through Python and C++ APIs. |
| Optical fiber cables | n.a. | Fiber-optic cables used for laser delivery and optical signal routing between system components. |
| USB cables and power supplies | n.a. | Connection and power accessories required for operation of the supplied hardware components. |

The pulsed laser module provides the optical excitation signal. The laser output is routed through a fiber connection to the experimental optical setup. Detected photons are converted into electrical pulses by the SPAD detector.

The detector output and laser synchronization signal are connected to the FELIX Time-to-Digital Converter. FELIX records photon arrival times and supports time-tagging as well as real-time hardware histogramming.

The timing data are transferred to a connected computer through USB and processed by the supplied software.

## 2.5 System Requirements

The FELIX Time-to-Digital Converter requires a compatible computer for configuration, acquisition, and operation. The computer shall satisfy the minimum requirements specified by the TEDIEL software documentation.

- **OS:** Windows 10 or Ubuntu 20.04 (or later), 64-bit only
- **CPU:** Intel or AMD processor with 64-bit support
- **RAM:** 4GB; recommended 12 GB or more for high performance time-tagging
- **Disk Storage:** 2 GB of free disk space
- **Monitor Resolution:** 1920×1080 minimum

An available USB interface compatible with the FELIX Time-to-Digital Converter is required.

Refer to the latest software documentation for updated system requirements.

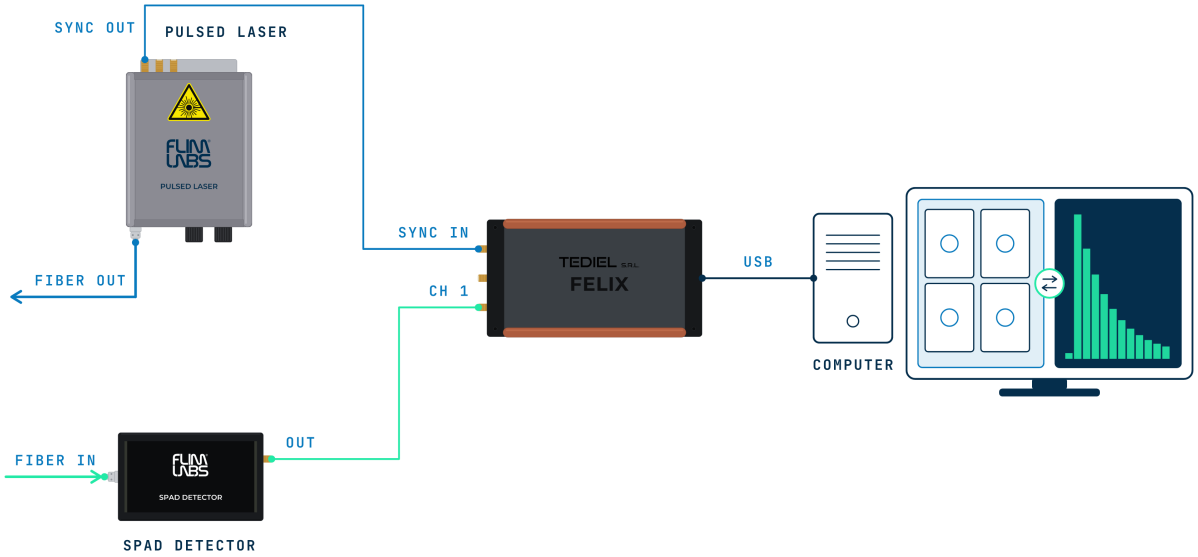



**Figure 2.1:** System architecture of the Quantum Photonics Bundle.

# 3 Safety Information

This chapter describes general safety instructions and residual risks associated with the Quantum Photonics Bundle.

## 3.1 General Safety Instructions

Before installation and operation, ensure that this manual and the documentation of all supplied components are complete and legible. If any part of the documentation is missing or unclear, contact the respective manufacturer before proceeding.

The bundle must be used only as described in this manual. Any operation not explicitly described in this manual is considered improper use.



**Warning** Before performing any installation or electrical connection procedure, disconnect power from all components unless explicitly instructed otherwise.

## 3.2 Laser Safety

The bundle includes a pulsed laser source.

Depending on the selected configuration, the system may emit visible and/or invisible laser radiation. Direct exposure to laser radiation may result in eye injury.



**Caution** Visible and invisible laser radiation.
Avoid direct eye exposure to laser radiation.
Do not disconnect optical fibers while the laser source is active.
Operate the laser only under safe operating conditions and according to all supplied documentation.

## 3.3 Electrical Safety

The bundle contains electronic devices intended for laboratory use.

- Operate all components only with the specified power supplies and interfaces.
- Use only undamaged and properly rated cables.
- Disconnect power before modifying electrical connections.
- Do not open the enclosures of electronic devices.
- There are no user-serviceable internal components.
- Keep all electronic components away from water sources and other liquids.
- Ensure that cables are routed to avoid mechanical damage and accidental disconnection.

## 3.4 Optical Safety and Fiber Handling

Optical fibers and connectors must be handled carefully to avoid damage, contamination, or unintended exposure to optical radiation.

- Keep fiber connectors clean and protected with dust caps when not in use.
- Do not bend optical fibers below their specified minimum bending radius.
- Do not inspect active fiber outputs directly.
- Ensure that all fiber connections are secure before enabling the laser.

## 3.5 Environmental and Operating Conditions

The Quantum Photonics Bundleis designed for indoor laboratory use only.
Typical operating conditions include:

- Indoor laboratory environment
- Stable, vibration-free installation surface

- Non-condensing humidity conditions
- Adequate ventilation around electronic components
- Protection from liquids, dust, and mechanical impact

Do not operate the system in explosive atmospheres or in the presence of flammable gases.

## 3.6 Modification and Improper Use

The supplied components must not be modified, altered, or repaired by unauthorized personnel.

Unauthorized modification of electrical connections, optical connections, laser safety features, or electronic devices may result in hazardous operating conditions, incorrect measurements, equipment damage, or loss of regulatory compliance.

# 4 Setup

This chapter describes the correct procedure for unpacking, installing, connecting, and verifying the Quantum Photonics Bundlebefore operation.

## 4.1 Scope of Delivery

The exact scope of delivery depends on the ordered system configuration.

**Table 4.1:** Bundle configuration and scope of delivery

| Component | Quantity |
|---|---|
| Picosecond Pulsed Laser Module | 1 |
| Single-Photon SPAD Detector | 1 |
| FELIX Time-to-Digital Converter | 1 |
| TEDIEL Software Suite | 1 |
| Optical fiber cables | As configured |
| USB cables and power supplies | 1 set |

Verify that all required items are present before proceeding with installation.

## 4.2 Unpacking and Inspection

The components of the bundle are supplied in protective packaging intended to protect them during storage and transport.

After unpacking:

- Inspect all components and accessories for visible damage.
- Check for dents, deformation, cracks, or signs of impact.
- Verify that all connectors and cables are intact.
- Inspect optical connectors and optical surfaces for contamination or damage.
- Verify that all ordered accessories are present.

If any damage, missing parts, or abnormalities are detected, stop the installation and contact the manufacturer before proceeding.



**Warning** Keep packaging materials out of reach of children and dispose of them according to local regulations.

## 4.3 Installation Requirements

The installation area must provide:

- Adequate workspace around the optical setup
- Stable laboratory surfaces
- Adequate ventilation for electronic components
- Protection from liquids and excessive dust
- Safe cable routing without tripping hazards
- Suitable laser safety measures for the selected configuration

## 4.4 Hardware Installation

### 4.4.1 Mechanical Installation

Figure 2.1 shows an overview of all system components and their connections. Install the supplied components in locations that allow safe cable routing and stable operation.

Ensure that:

- Electronic devices are placed on stable surfaces.
- Components are protected from mechanical stress and accidental impact.
- Adequate space is available for ventilation and cable routing.
- Optical fibers and cables are not bent below their minimum bending radius.

### 4.4.2 Optical Connections

Connect the pulsed laser module to the optical setup using the supplied optical fiber. Connect the SPAD detector to the optical detection path according to the experimental configuration.

Before connecting optical fibers ensure that:

- Optical connectors are clean before connection.
- Fiber connectors are correctly aligned before tightening.
- Optical fibers are protected from excessive bending and mechanical stress.
- Dust caps are removed only immediately before connection.

### 4.4.3 Signal and Data Connections

Connect the laser synchronization signal and detector output signal to the FELIX Time-to-Digital Converter according to the system configuration.

Connect the FELIX Time-to-Digital Converter to the computer using the supplied USB cable.

### 4.4.4 Power Supply

Connect all devices only to compatible power supplies and USB interfaces.

Do not apply power to the system until all optical and electrical connections have been completed and verified.

## 4.5 Software Installation and Initial Verification

Data acquisition and analysis are performed using the supplied software.

After completing the hardware installation and software installation:

1. Verify that all optical and electrical connections are complete.
2. Power on the connected computer.
3. Power on all supplied hardware components.
4. Start the supplied software.
5. Verify that the FELIX Time-to-Digital Converter is detected.
6. Verify that detector and synchronization signals are present.
7. Configure the acquisition settings according to the experiment.
8. Perform a test acquisition before starting the final measurement.

# 5 Operation

This chapter describes the typical workflow for operating the Quantum Photonics Bundle.

## 5.1 System Operation Principle

During operation, the pulsed laser generates optical excitation pulses. The detector records single photons generated or transmitted by the experimental setup. The FELIX Time-to-Digital Converter measures photon arrival times relative to the selected timing reference.

The software displays, records, and processes the acquired timing information.

## 5.2 Startup Procedure

A typical startup workflow is listed below:

1. Verify that all optical and electrical connections are complete.
2. Ensure that laser safety precautions are in place.
3. Power on the connected computer.
4. Power on all supplied hardware components.
5. Start the supplied software.
6. Load or create the required acquisition configuration.
7. Verify that the FELIX Time-to-Digital Converter is detected correctly.
8. Verify that detector and synchronization signals are present.
9. Start acquisition.

## 5.3 Acquisition Workflow

Typical measurements may include:

- Photon counting
- Time-tagging
- Photon correlation measurements
- Real-time hardware histogramming
- Time-resolved decay measurements

Detailed descriptions of software modes, acquisition settings, APIs, and analysis functions are provided in the dedicated software documentation.

## 5.4 Data Export

The supplied software supports export of acquired timing and histogram data for further analysis and storage.

Users should verify the selected export location, file name, and acquisition settings before starting long measurements.



**Notice** Verify that data export is configured correctly before starting long acquisitions to avoid unintended data loss.

## 5.5 Shutdown Procedure

After completing measurements:

1. Stop acquisition in the software.
2. Save and export all required measurement data.
3. Disable the laser output.
4. Close the software.
5. Power down the supplied hardware components.

6. Disconnect power only after the system has been safely shut down.

## 5.6 Troubleshooting

**Table 5.1:** Common issues, possible causes, and solutions

| Issue | Possible Cause | Solution |
|---|---|---|
| No hardware detected | USB connection or driver issue. | Check the USB connection, restart the software, and reinstall the driver if required. |
| No synchronization signal detected | Incorrect sync connection or trigger configuration. | Check the synchronization cable, signal source, and trigger settings. |
| No photon counts detected | Detector signal missing, laser inactive, or incorrect input selected. | Check detector output, laser status, input channel selection, and cable connection. |
| Unexpected count rate | Incorrect optical alignment, excessive background light, or detector saturation. | Check optical alignment, reduce background light, and adjust laser power or attenuation. |
| Unstable acquisition | USB bandwidth or connection issue. | Use a direct USB connection and avoid unsupported USB hubs. |

# 6 Maintenance and Service

This chapter describes the cleaning, preventive maintenance, corrective maintenance, and authorized service procedures for the supplied hardware components. All maintenance activities must be performed by qualified personnel who have read and understood this manual.

Maintenance must be carried out in accordance with this manual and applicable workplace safety regulations. The maintenance area should provide adequate illumination; a minimum lighting level of 200 lux is recommended.

Before performing any cleaning or maintenance operation, switch off all powered components and disconnect them from their power sources.

## 6.1 Cleaning

The system components require periodic external cleaning to ensure reliable operation.

- Remove dust or dirt from exterior surfaces using a dry or slightly damp lint-free cloth.
- A mild, neutral detergent may be used if necessary.
- Do not allow liquids to enter connectors, optical interfaces, or electronic components.
- Avoid the use of solvents, aggressive chemicals, or abrasive materials.

For connectors and hard-to-reach areas, such as USB ports or ventilation openings, a soft-bristled brush or compressed air may be used.

### Optical Port and Connector Cleaning

Proper cleaning of optical connectors is essential to ensure reliable optical coupling and to prevent signal attenuation or permanent damage to optical interfaces.

Always inspect the ferrule end faces of fiber patch cables and optical bulkhead connectors before making optical connections. Ferrule end faces may be cleaned using commercially available lint-free fiber cleaning wipes or by applying a small amount of isopropanol (IPA) to a clean lint-free cloth.

Optical bulkhead connectors are best cleaned using a dedicated fiber-optic cleaning pen or cleaning stick designed for the connector type in use. Follow the instructions provided by the manufacturer of the cleaning tool.



**Caution** Before cleaning any optical connector or fiber interface, ensure that all laser sources are switched off and that no optical radiation is present in the system.

## 6.2 Preventive Maintenance

The supplied FLIM LABS hardware components do not require routine internal maintenance.

Periodically inspect:

- External surfaces for mechanical damage
- Connectors and cables for wear, deformation, or contamination
- Optical connectors and fiber patch cables for cleanliness and damage
- Product labels for legibility

If any abnormal condition is detected, discontinue use of the affected component and contact technical support.

## 6.3 Service and Repairs

The supplied hardware components contain no user-serviceable internal parts. In the event of malfunction, accidental damage, or improper operation:

- Switch off and disconnect the affected component.
- Do not attempt internal repair.

- Contact the manufacturer’s technical support for instructions.

Repairs or modifications must be carried out exclusively by FLIM LABS S.r.l. or authorized personnel. Use only original FLIM LABS spare parts and approved replacement accessories. Further warranty conditions are described in Section 1.6.



Unauthorized intervention or the use of non-original components may compromise safety, performance, and warranty coverage.

# 7 Decommissioning and Disposal

This chapter describes procedures for removing system components from service and disposing of them in accordance with applicable regulations.

## 7.1 Decommissioning Procedure

The supplied hardware components are designed for long-term laboratory use. When a component reaches the end of its technical or operational life, it must be decommissioned to prevent further use.

Decommissioning requires:

- Disconnecting the component from all power sources.
- Removing all signal, communication, and optical connections.
- Ensuring that the component cannot be unintentionally reused.

The same procedure applies in cases of:

- Permanent withdrawal from service
- Storage without intended future use
- Final disposal

Reuse of components after final decommissioning is not permitted unless explicitly authorized by the manufacturer.

## 7.2 Disposal

All hardware components supplied with the system must be disposed of in accordance with applicable local and European regulations governing electrical and electronic equipment.

In accordance with Directive 2012/19/EU (WEEE), electronic components must not be disposed of with household waste. The product labels include the WEEE symbol indicating the requirement for separate collection of electrical and electronic equipment. Contact your local waste management authority or a certified collection center for proper disposal procedures.

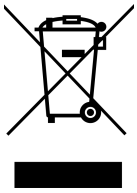


**Figure 7.1:** WEEE symbol